STOCK MARKET ANALYSIS USING SQL

Insights Report | Six NSE Stocks | January 2015 – July 2018

1. Objective and approach

The objective of this project is to use SQL to explore historical closing prices for Bajaj Auto, TCS, TVS Motors, Hero MotoCorp, Infosys and Eicher Motors; calculate moving averages and crossover signals; compare price performance; and investigate unusually large daily price changes. The analysis uses closing prices and SQL queries, including aggregations, common table expressions (CTEs), joins, CASE expressions and window functions.

2. Full-period price performance (Task 11)

Stock	First close (₹)	Last close (₹)	Change
TVS Motors	276.85	517.45	+86.9%
Eicher Motors	15,239.15	27,820.95	+82.6%
Bajaj Auto	2,454.10	2,700.70	+10.0%
Hero MotoCorp	3,107.30	3,293.80	+6.0%
TCS	2,548.20	1,941.25	–23.8%
Infosys	1,975.80	1,365.00	–30.9%

Finding: TVS Motors (+86.9%) and Eicher Motors (+82.6%) had the strongest raw price growth. Bajaj Auto (+10.0%) and Hero MotoCorp (+6.0%) had smaller gains. TCS (–23.8%) and Infosys (–30.9%) ended below their starting prices in the unadjusted comparison. Because bonus issues affect quoted prices, TCS and Infosys should also be assessed using adjusted prices (Task 13).

3. Moving-average crossover signals (Tasks 5–10)

The 20-day moving average responds more quickly to recent price movements, while the 50-day moving average is slower and reflects a longer trend. A Buy signal is generated when the 20-day average crosses above the 50-day average; a Sell signal is generated when it crosses below; otherwise the signal is Hold. These indicators use past prices only, so a crossover confirms a change after some movement has already occurred and may lag a turning point.

Stock	Buy	Sell	Latest recorded non-Hold signal
Bajaj Auto	12	12	2018-06-21 — Buy
Eicher Motors	7	7	2018-06-06 — Sell
Hero MotoCorp	9	9	2018-05-22 — Sell
Infosys	10	9	2018-05-07 — Sell
TCS	13	14	2018-06-05 — Buy
TVS Motors	9	9	2018-05-17 — Sell
Total	**60**	**61**	**—**

Interpretation: The recorded signal output shows the latest non-Hold signal as Buy for Bajaj Auto and TCS, and Sell for Eicher Motors, Hero MotoCorp, Infosys and TVS Motors. These signals are not standalone investment advice; they should be considered alongside the long-run price trend and corporate-action adjustments.

> **Signal-count checkpoint:** The recorded Task 10 query output totals 60 Buy and 61 Sell signals, whereas the student guide checkpoint lists 56 Buy and 57 Sell. This discrepancy remains unresolved in the recorded results and should be checked against the final SQL before relying on the aggregate counts.

Signal whipsaws and transaction cost: moving-average crossovers can reverse during sideways or volatile markets. This project output does not include a verified count of rapid Buy/Sell reversals or a transaction-by-transaction backtest, so the monetary cost of following every signal cannot be quantified from the results recorded here. Brokerage, taxes, slippage and execution prices would also be required for a realistic estimate.

4. Largest daily declines and corporate actions (Task 12)

Stock	Date	Close (■)	Daily move
TCS	2018-05-31	1,744.80	–50.4%
Infosys	2015-06-15	991.10	–49.9%
TVS Motors	2016-05-03	287.90	–9.7%
Eicher Motors	2015-08-24	18,571.75	–9.3%
Bajaj Auto	2015-08-24	2,188.45	–9.1%
Hero MotoCorp	2018-07-23	3,163.90	–6.2%

The two extreme moves are TCS on 31 May 2018 (–50.4%) and Infosys on 15 June 2015 (–49.9%). These dates align with 1:1 bonus ex-dates reported in corporate-market records. In a 1:1 bonus issue, the share count doubles and the quoted price may approximately halve; the raw price drop therefore should not be read as an equivalent loss of shareholder value. The other stocks' largest reported daily declines are approximately 6%–10%.

5. Bonus-adjusted price comparison (Task 13)

To compare performance across each bonus event, the closing prices before the event date are divided by 2, while prices on and after the event date are left unchanged. The SQL adjustment uses the first date trading at the new price level.

Stock	Adjustment applied	Adjusted full-period change
TCS	Prices before 2018-05-31 ÷ 2	+52.4%
Infosys	Prices before 2015-06-15 ÷ 2	+38.2%

After the bonus adjustment, TCS changes from an apparent raw decline (–23.8%) to an adjusted gain (+52.4%). Infosys changes from an apparent raw decline (–30.9%) to an adjusted gain (+38.2%). This demonstrates why corporate actions must be accounted for before comparing historical price performance. These calculations adjust for the identified bonus events only; they are not total shareholder returns because dividends and other factors are not included.

6. Overall conclusion and limitations

SQL made it possible to summarise trading history, compare prices, calculate 20-day and 50-day moving averages, generate crossover signals and identify abnormal daily price changes. TVS Motors and Eicher Motors had the largest raw price gains; after accounting for the identified 1:1 bonus issues, TCS and Infosys also show positive adjusted price changes over the period. The latest recorded crossover signals are mixed across the six companies and should not be treated as guaranteed predictions.

Limitations: The dataset and queries do not include dividends, brokerage, taxes, slippage, company announcements, fundamental financials or broader market conditions. Moving averages are backward-looking and may generate delayed or false signals. Exact rapid-reversal costs were not calculated. In addition, the Task 10 aggregate signal count differs from the guide checkpoint, so that SQL result requires a final check.

References for corporate-action dates

TCS: TCS investor-relations bonus issue information and the TCS 2018 buyback letter of offer identify the 1:1 bonus ex-date as 31 May 2018.
Infosys: Infosys corporate-market bonus history and investor-relations information identify the 1:1 bonus event on 15 June 2015.

Sources: TCS Investor Relations, “Issue of Bonus Shares – 2018” and “Investor FAQs”; Metropolitan Stock Exchange of India, corporate securities information for TCS and Infosys; Infosys Investor Relations, quarterly report for Q1 FY2015–16.